EQ Credit: Thermal Comfort

This credit applies to

- BD+C: New Construction (1 point)
- BD+C: Schools (1 point)
- BD+C: Retail (1 point)
- BD+C: Data Centers (1 point)
- BD+C: Warehouses & Distribution Centers (1 point)
- BD+C: Hospitality (1 point)
- BD+C: Healthcare (1 point)

INTENT
To promote occupants' productivity, comfort, and well-being by providing quality thermal comfort.

REQUIREMENTS
Meet the requirements for both thermal comfort design and thermal comfort control.

Thermal Comfort Design

NC, SCHOOLS, RETAIL, DATA CENTERS, HOSPITALITY, HEALTHCARE

Design heating, ventilating, and air-conditioning (HVAC) systems and the building envelope to meet the requirements of ASHRAE Standard 55–2017, Thermal Comfort Conditions for Human Occupancy with errata or a local equivalent.

For natatoriums, demonstrate compliance with ASHRAE HVAC Applications Handbook, 2015 edition, Chapter 5, Places of Assembly, Typical Natatorium Design Conditions, with errata.

Data Centers only
Meet the above requirements for regularly occupied spaces.

WAREHOUSES & DISTRIBUTION CENTERS
Meet the above requirements for office portions of the building.

In regularly occupied areas of the building's bulk storage, sorting, and distribution areas, include one or more of the following design alternatives:

- radiant flooring;
- circulating fans;
- passive systems, such as nighttime air, heat venting, or wind flow;
- localized active cooling (refrigerant or evaporative-based systems) or heating systems; and
- localized, hard-wired fans that provide air movement for occupants' comfort.
- other equivalent thermal comfort strategy.

Thermal Comfort Control

NC, SCHOOLS, RETAIL, DATA CENTERS, WAREHOUSES & DISTRIBUTION CENTERS, HOSPITALITY

Provide individual thermal comfort controls for at least 50% of individual occupant spaces. Provide group thermal comfort controls for all shared multioccupant spaces.

Thermal comfort controls allow occupants, whether in individual spaces or shared multioccupant spaces, to adjust at least one of the following in their local environment: air temperature, radiant temperature, air speed, and humidity.

Hospitality only
Guest rooms are assumed to provide adequate thermal comfort controls and are therefore not included in the credit calculations.

Retail only
Meet the above requirements for at least 50% of the individual occupant spaces in office and administrative areas.

HEALTHCARE

Provide individual thermal comfort controls for every patient room and at least 50% of the remaining individual occupant spaces. Provide group thermal comfort controls for all shared multioccupant spaces.

Thermal comfort controls allow occupants, whether in individual spaces or shared multioccupant spaces, to adjust at least one of the following in their local environment: air temperature, radiant temperature, air speed, and humidity.

GUIDANCE

Refer to the LEED v4 reference guide, with the following additions and modifications:

Behind the Intent

Beta Update

The ASHRAE standards have been updated to the latest versions:

- ASHRAE 55-2010 → ASHRAE 55-2017 (updated version)
  - Major changes from 2010 include: clarifications to elevated air speed method, new requirement to calculate the change to thermal comfort resulting from direct solar radiation impacting occupants.
- ASHRAE Applications Handbook 2011 edition → ASHRAE Applications Handbook 2015 edition (updated version)

Option 2 is available for use through regional compliance pathways.

Step-by-Step Guidance

Step 1. Select Analysis Method(s)

Select the methodology from ASHRAE Standard 55-2017 that will be used for the thermal comfort analysis. Refer to the standard for restrictions with some of the methods. For example, section 5.3.1 is restricted to conditions without direct solar radiation.

Additional calculations or considerations required if direct-beam solar radiation falls on the occupant. Calculations are available in ASHRAE 55-2017 Normative Appendix C.

Tools such as the CBE Thermal Comfort Tool (https://comfort.cbe.berkeley.edu/) may be used to assist with the analysis. A new feature (SolarCal) is also available for the direct-beam solar radiation calculations.

Step 2. Design Project's Conditioning Systems

Design the project's conditioning systems to provide the acceptable comfort conditions identified in the analysis. Verify that all spaces at risk for discomfort, such as locations close to the entrances prone to drafts or west-facing walls that may retain heat, have been addressed.

ASHRAE 55–2017, Section 6.1, requires the design to be within the acceptable comfort range at all combinations of conditions that are expected to occur, including variations in internal loads and the exterior environment, and at both full- and partial-load conditions. Systems that cannot maintain comfort under all conditions (e.g., a constant-volume rooftop unit with a single compressor may have problems controlling humidity levels) do not meet the credit requirements.

Further Explanation

Criteria for Occupant-Controlled Naturally Conditioned Spaces

The same set of requirements per LEED v4 reference guide for use of occupant controlled naturally conditioned spaces (or adaptive) method are applicable for the newer standards ASHRAE 55-2017 (Section 5.4) and ISO 17772-201 (Section H.2) as well.

Examples

Example 1: Follow guidance per LEED v4 reference guide with the exception of referring to ASHRAE 55-2017, Table 5.2.2.B for garment insulation values and Graphic Comfort Zone Method per Section 5.3.1

Example 2: Follow guidance per LEED v4 reference guide with the exception of referring to ASHRAE 55-2017, Table 5.2.1.2 and related Appendix F as well as Analytical Comfort Zone Method per Section 5.3.2 that incorporates the computer model method.

Example 3: Follow guidance per LEED v4 reference guide with the exception of referring to ASHRAE 55-2017, Section 5.4, Method for Determining Acceptable Conditions in Naturally Conditioned Spaces and plotting the average monthly outdoor temperatures and design operative temperatures per Figure 5.4.2.

Example 4: Follow guidance per LEED v4 reference guide with the exception of referring to ISO 17772-2017, Section 6.2.2 and Figure H.1- Default Design Values for the Indoor Operative Temperature for Buildings without Mechanical Cooling for using the adaptive method.

Project Type Variations

Refer to LEED v4 reference guide with the following modifications:

Use ASHRAE 55-2017, Appendix F for guidance on Gymnasiums, Fitness areas and other spaces with high metabolic rates

Natatoriums: Use ASHRAE HVAC Applications handbook, 2015 edition for typical natatorium design conditions guidance.

Required Documentation

Refer to LEED v4 ref guide with the following modifications:

- For thermal comfort design compliance, supporting documentation verifying that the thermally conditioned spaces meet ASHRAE Standard 55-2017 for 80% acceptability. Examples include a psychometric chart, PMV/PPD calculations, CBE Thermal Comfort Tool results, a copy of ASHRAE Standard 55-2017 Figure I2, Figure I4, and/or Figure I5, or predicted worst case (both heating and cooling) indoor conditions for each month on a copy of ASHRAE Standard 55-2017 Figure 5.4.2.
- Explanation of how project complies with the direct solar radiation requirements in ASHRAE Standard 55-2017.
    - Screenshots of the local discomfort assessment and SolarCal windows from the CBE Thermal Comfort Tool are acceptable.

Connection with Ongoing Building Performance

- LEED O+M EQ prerequisite Indoor Environmental Quality Performance: Strategies to promote occupant comfort and wellbeing by providing thermal comfort such as designing HVAC systems and building envelope per thermal comfort standards and providing individual thermal comfort controls in newly constructed occupant spaces can contribute to better indoor environmental quality and overall occupant satisfaction during operations phase.